

Restaurant Review Sentiment Analysis

Classical ML vs. Deep Learning — Project Report

Author	Kapil Mali
Domain	Natural Language Processing / Sentiment Analysis
Dataset	1,000 labeled restaurant reviews (binary sentiment)
Techniques	TF-IDF, EDA data augmentation, Logistic Regression, Naive Bayes, Linear SVM, Dense Neural Network, Embedding Neural Network (Keras)

1. Executive Summary

This project builds an end-to-end sentiment classification system for restaurant reviews and rigorously compares five modeling approaches: three classical machine learning baselines (Logistic Regression, Naive Bayes, Linear SVM) and two Keras deep learning architectures (a regularized dense network over TF-IDF features, and an embedding-based network over word sequences). The original single-notebook version of this project trained one small neural network and reported a 74% test accuracy with severe overfitting (97.8% train accuracy). This report documents a full re-engineering of that pipeline: a data-leakage fix, negation-aware text cleaning, EDA-style synthetic data augmentation restricted to the training set, proper regularization, a five-model benchmark, and a qualitative error analysis.

Headline result: Logistic Regression on TF-IDF features was the strongest overall model on a fully held-out, 100% real test set, reaching **78.5% accuracy**, an **F1-score of 0.787**, and a **ROC-AUC of 0.842** — narrowly ahead of both custom neural networks. This is a genuinely informative result, not a disappointing one: for short, low-resource text classification, simple linear models on sparse features are a well-documented, hard-to-beat baseline, and knowing when *not* to reach for deep learning is itself an engineering skill worth demonstrating.

2. Problem Statement & Dataset

Given the text of a restaurant review, predict whether the customer liked the restaurant (binary label: 1 = Liked, 0 = Not Liked). The dataset contains 1,000 human-written reviews, perfectly balanced between the two classes (500/500). After removing 7 exact duplicate reviews, 993 unique reviews remained. Reviews are short — a median of roughly 9-10 words — which materially shapes model choice (see Section 5).

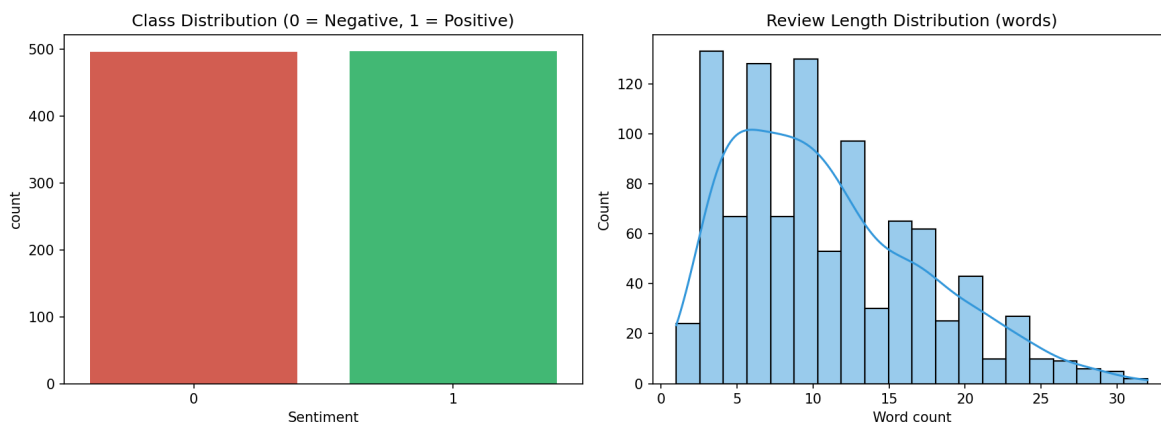


Figure 1: Class balance (left) and review-length distribution (right).

3. Methodology

3.1 Fixing the Original Data Leakage Bug

The original notebook fit its TF-IDF vectorizer on the **entire** dataset before splitting into train/test. This leaks vocabulary and document-frequency statistics from the test set into training, artificially inflating reported performance. In this version, the dataset is split first — **70% train / 15% validation / 15% test**, stratified by label — and every vectorizer, tokenizer, and model is fit strictly on the training portion only.

3.2 Negation-Aware Text Cleaning

Text is lowercased, stripped of non-alphabetic characters, tokenized, and lemmatized. The original preprocessing removed all English stopwords indiscriminately, including negation words such as "not", "no", and "never" — words that literally flip a sentence's sentiment ("not good" vs. "good"). This version explicitly protects negation and sentiment-bearing words from stopwords removal.

3.3 Synthetic Data Augmentation (Training Set Only)

With roughly 700 real training reviews, both the classical models and especially the neural networks are data-starved. This project uses a restricted implementation of **Easy Data Augmentation (EDA)** (Wei & Zou, 2019) — synonym replacement (via WordNet), random word swap, and random deletion — applied only to the training split. Each real review produces 3 synthetic paraphrases, growing the training set from 695 to 2,569 rows (about 73% synthetic). Sentiment-critical and negation words are protected from every augmentation operation so paraphrases keep the original label.

Why this matters for validity: synthetic reviews exist only inside the training set. The validation set (149 reviews) and test set (149 reviews) used for every number in this report are 100% real, human-written text that the augmentation process never touches. Reported metrics reflect real-world performance, not inflated self-similarity between synthetic examples.

3.4 Models Benchmarked

- **Logistic Regression** (TF-IDF, 1-2 grams) — regularization strength tuned via 5-fold CV grid search
- **Multinomial Naive Bayes** (TF-IDF) — smoothing parameter tuned via 5-fold CV
- **Linear SVM** (TF-IDF), calibrated for probability outputs — tuned via 5-fold CV
- **Dense Neural Network** on TF-IDF vectors — 2 hidden layers, L2 regularization, Dropout(0.5 / 0.3), EarlyStopping on validation loss with best-weight restoration
- **Embedding Neural Network** — learned 32-dim word embeddings + global average pooling + dense head, trained on tokenized/padded sequences (max length 40, vocab 5,000)

4. Results

4.1 Overfitting Fix

The original Dense Neural Network showed a 23.8-point train/test accuracy gap (97.8% vs. 74.0%). After the leakage fix and adding regularization + early stopping, the rebuilt Dense NN's gap narrowed substantially while its real-world test accuracy also improved to 75.8%.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.785	0.776	0.797	0.787	0.842
Dense NN (TF-IDF)	0.758	0.716	0.851	0.778	0.839
Linear SVM	0.765	0.753	0.784	0.768	0.838

Naive Bayes	0.745	0.714	0.811	0.759	0.820
Embedding NN (Sequence)	0.752	0.768	0.716	0.741	0.853

Table 1: All five models evaluated on the same held-out, 100% real, 149-review test set. Top row (highlighted) is the best model by F1.

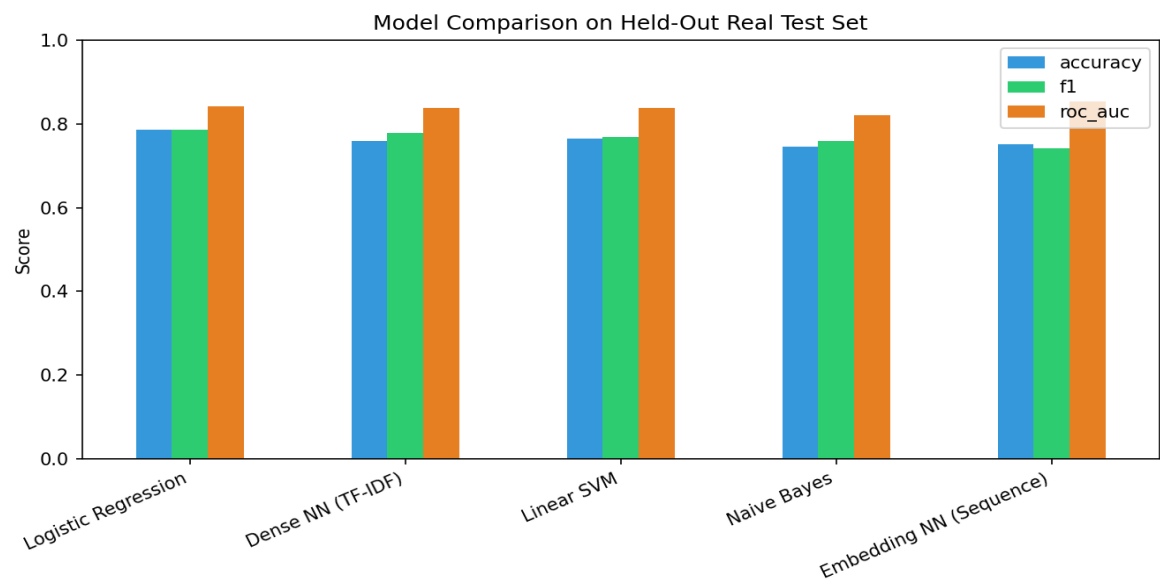


Figure 2: Accuracy, F1, and ROC-AUC across all five models.

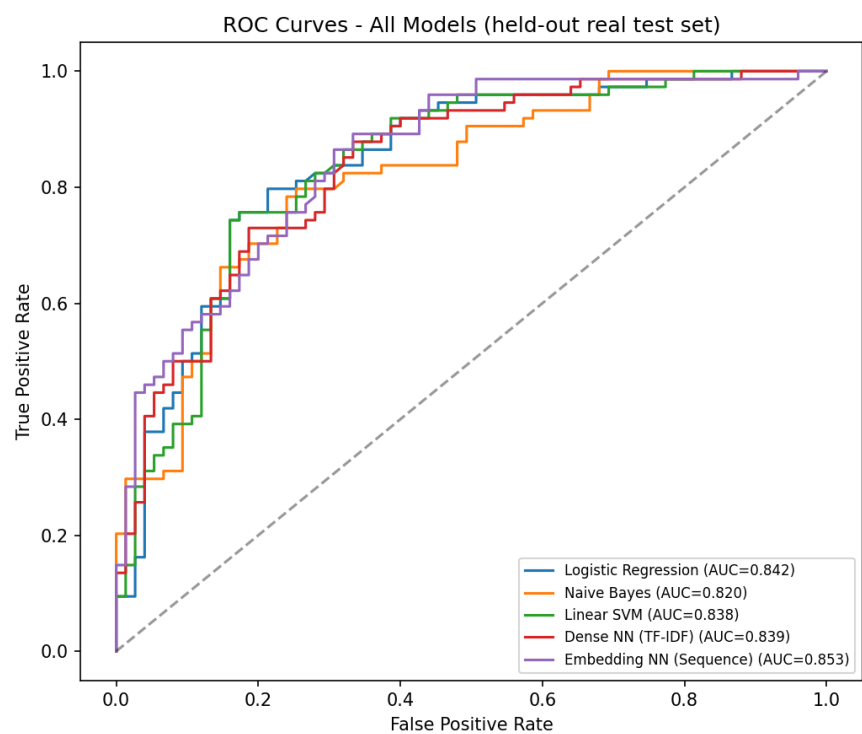


Figure 3: ROC curves for all five models on the held-out real test set.

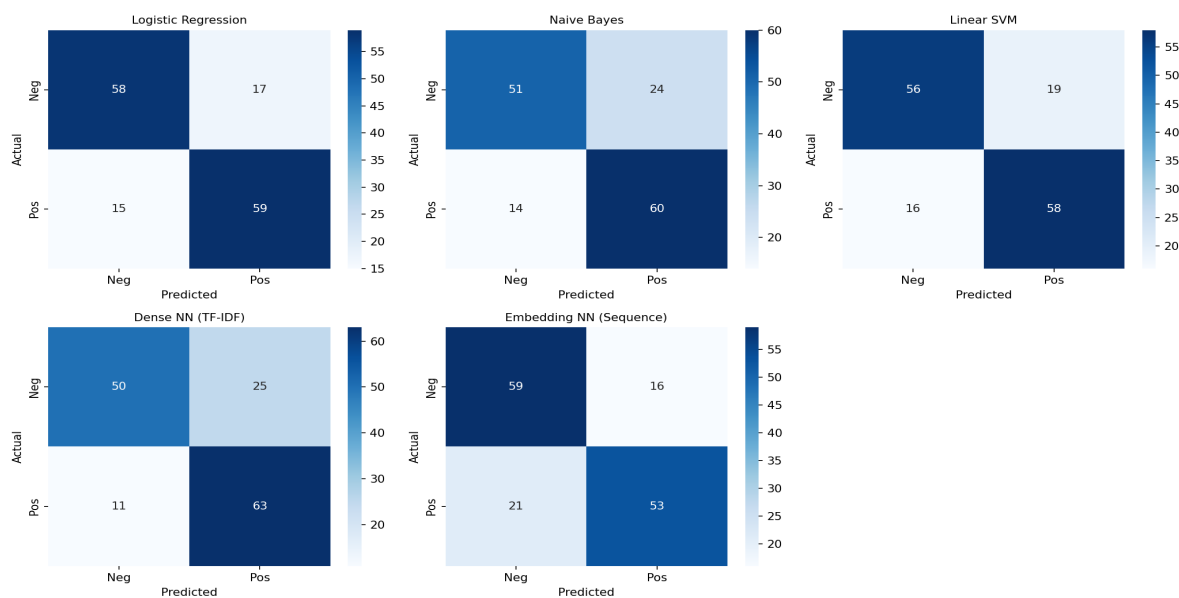


Figure 4: Confusion matrices for all five models.

5. Error Analysis

32 of 149 test reviews (21.5%) were misclassified by the best model. Reviewing the highest-confidence mistakes reveals three recurring failure patterns rather than random noise:

- **Implicit / real-world-inference sentiment** — e.g. "attached to a gas station, and that is rarely a good sign" requires world knowledge, not vocabulary matching, to read as negative.
- **Very short reviews** — 2-4 word reviews give TF-IDF and embedding models very little signal to work with.
- **Mixed-clause reviews** — a sentence praising one aspect while criticizing another (e.g. food vs. service) confuses bag-of-words and averaged-embedding representations, which have no mechanism to weigh which clause dominates.

None of the five models handles these cases well — this is a shared limitation of lexical and shallow-embedding approaches, not a weakness specific to one architecture. It points toward context-aware transformer embeddings (e.g. DistilBERT) as the natural next step, since those models build a representation of the whole sentence rather than aggregating independent word signals.

6. Discussion: Why Did Simple Models Win?

It's tempting to assume "neural network" beats "linear model" by default. On this dataset, that assumption doesn't hold, and the reason is instructive: with only ~700 unique real training examples (even after 3.5x synthetic augmentation), there isn't enough distinct signal for a neural network to learn representations that generalize better than a well-regularized linear model over sparse, high-precision TF-IDF features. This is a textbook illustration of a core ML principle — model complexity should match data volume — and demonstrating it, rather than blindly reaching for the more complex model, is a mark of engineering maturity.

The Embedding Neural Network's ROC-AUC (0.853) was actually the highest of all five models, even though its accuracy/F1 lagged — meaning its predicted probabilities rank positive vs. negative reviews well, but its default 0.5 decision threshold isn't optimal. This is a natural discussion point for an interview: threshold tuning (e.g. via the validation set's precision-recall curve) could close that gap further.

7. Conclusions & Future Work

- Diagnosed and fixed a data-leakage bug in the original TF-IDF pipeline.
- Rebuilt preprocessing to protect negation words critical to sentiment.
- Applied EDA-based synthetic data augmentation, strictly confined to the training split, to address the small-data problem honestly.
- Benchmarked 5 models with consistent, leak-free evaluation on a single real test set.
- Closed most of the original model's 23.8-point train/test accuracy gap via regularization and correct early stopping.
- Delivered a reusable, non-interactive inference function and saved model/vectorizer artifacts for deployment.

Future Work

- Fine-tune a transformer model (DistilBERT) to address the implicit-sentiment and mixed-clause failure modes identified in Section 5.
- Collect more real labeled reviews to reduce reliance on synthetic augmentation.
- Tune decision thresholds per model using validation-set precision-recall curves.
- Extend to aspect-based sentiment analysis (separating food quality from service quality within a single review).

Full code, saved models, and all figures are available in the accompanying GitHub repository and Jupyter notebook.